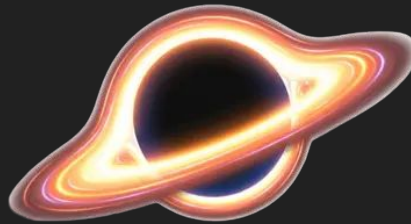


Science in Film

Alexander Friesen





Which movie does NASA use
to train new employees?

A. Gravity

C. Armageddon

B. Interstellar

D. Star Wars



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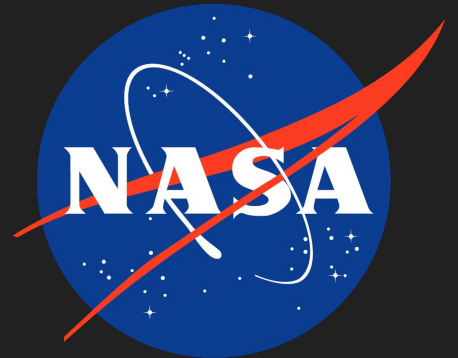
D. Star Wars

ARMAGEDDON

has over

168

scientific inaccuracies –



Why science in film?

- “An Astrophysicist Goes to the Movies” show
- Critiqued various films, TV shows, and beer commercials
- Inspired to research Science in Film



Physics in
Space

The Passage
of Time

Genetic
Engineering

Robotics
and AI

Extradimensional
Physics

ARMAGEDDON

GRAVITY

BLACK HOLE

STAR TREK

STAR
WARS

THE
MARTIAN

THE TIME
MACHINE

BACK
TO THE FUTURE

PRIMER

LOOPER

TENET



GATTACA

AVATAR

PROJECT
HAIL
MARY

2001:
a space odyssey

METROPOLIS

i,ROBOT

MATRIX

INTERSTELLAR

INCEPTION

CONTACT

Research Findings

What are common
scientific **inaccuracies**
found in popular films?

Interstellar inaccuracies

- **Wormhole** appearance, timeframe, 5D beings
- **Hazardous objects** mile-high waves, frozen clouds
- **Blight** probability and impact



Armageddon **in**accuracies

- **Explosions** in space
- **Nuclear bomb** would have no impact
- **Texas-sized asteroid** 120x too big



What film ideas were
theoretical and now are
scientific **accuracies**?

Gattaca accuracies

- **Genetic Engineering** advancements in science
- **Bioethical** dilemmas
- **Epigenetic** influence on personality and outcomes



The Martian **accuracies**

- **Distance** from Earth
- **Toxicity** of Martian soil
- **Underground water** below surface



What scientific ideas could
inspire scientists if shown
in movies?

Interstellar **inspires**

- Stunning **theoretical depictions** of our universe
- Depicts **black holes**, wormholes, and **tesseract**s
- **Inspires future work** on extra dimensional objects



Tenet inspires

- **Time and entropy** of objects are **reversed**
- **Inverted** bullets, freezing fires, and time reversal
- **Inspires future work** on entropy and space time



Expert Interviews

Dr. Donna J. Nelson

- Chemistry advisor for *Breaking Bad*
- Reviewed diagrams and substances
- “Accuracy is always important in TV...”



From left to right: Bryan Cranston (Walter White), Dr. Donna J. Nelson, Aaron Paul (Jesse Pinkman).

Adam Savage

- *Mythbusters*
co-host, film special effects
- Star Wars, Matrix, Galaxy Quest
- “Almost nothing in Jaws is physically possible...”



Kip Thorne

- Executive producer of *Interstellar*, Nobel theoretical physicist
- Envisioned scientific scenes in *Interstellar*
- “When you’re going to spend some years working... love what you’re doing.”



Practical Application:

Science in Film Survey

Surveyed
1,034
US Citizens

- 57 Questions
- Demographically balanced
- 5 Scientific Areas

The General Public Perception of Science in Film

Alexander Friesen

Survey Responses and Analysis

Who scored science
questions more accurately?

A:

Science Degree

B:

Science Films

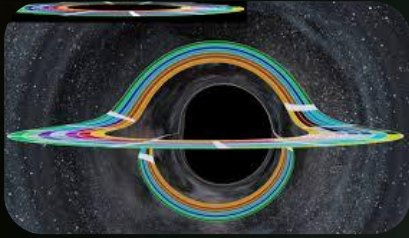
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Conclusion

Science and Art work best together



Science



Art

Acknowledgements

- Teachers (Brianna, Sylvia, Mike, Bonnie)
- Friends and Family (Paul, Larissa, Grandparents)
- Interviewees (Dr. Nelson, Adam Savage, Kip Thorne)
- Survey Respondents (all 1,034 people across the US)

Thank you!